

Falsifiability: How to Test the GTE Framework

The Complete Prediction and Falsification Register

novaspivack.com/research P48 complete theory monograph

doi.org/10.5281/zenodo.20669142 (*registration pending*)

UGP Physics Tutorial Series

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2026

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Prerequisites (read first): *The GTE Polynomial: A Step-by-Step Cheat Sheet · Perfect Self-Containment and MDL*

Next tutorials: None — this is a terminal reference tutorial.

See also: *From the Arithmetic Substrate to Particle Masses · The Weinberg Angle from $\mathbb{Z}_7$ Arithmetic · The Born Rule as a Theorem · New Physics Predictions: What GTE Predicts Will and Won't Be Found*

1 Why Falsifiability Matters for GTE

The core argument

A theory without free parameters is maximally falsifiable. GTE has zero continuously adjustable parameters: every numerical output is a derivation, not a fit. This makes GTE *more* falsifiable than the Standard Model, which has 19 free parameters (plus at least 3 for neutrinos) that can absorb discrepancies. A single confirmed measurement outside a GTE prediction band, at the stated precision, would refute the framework with no available escape route.

GTE (Generative Triple Evolution) derives all Standard Model parameters from a single 19-bit algebraic certificate: the polynomial $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$, selected by the **MDL (Minimum Description Length)** principle. With zero free parameters, every output of the framework is a prediction that could in principle be falsified.

This tutorial is the experimentalist's guide to GTE. It presents the complete prediction register organized by sector, identifies the strongest confirmations, explains the current tensions honestly, provides clean falsification criteria for each sector, and shows how each experimental community can contribute to testing the framework.

All comparisons use PDG 2024 [1] and NuFIT 6.0 IC24 NH [2] for neutrino parameters.

2 The Complete Prediction Register

2.1 Sector 1: Particle Masses

Nine charged-fermion masses spanning six orders of magnitude — three leptons and six quarks — are derived from the GTE arithmetic cascade with zero free parameters [1, 2].

The **Koide relation** $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ is a theorem (`ucl_lepton_sector_koide_identity`, **CatAL**), not a fit.

2.2 Sector 2: Mixing Angles and CP Violation

All three PMNS angles are derived from orbit-ratio formulas with zero free parameters (`pmns_solar_sin_sq`, `pmns_atm_sin_sq`, `pmns_reactor_sin_val`, all **CatAL**) [2].

2.3 Sector 3: Cosmology

The spectral tilt $n_s = 0.96488$ is derived from GTE's $\mathbb{Z}_7$ defect cosmology with zero parameters, machine-certified (`cmb_spectral_index_equals_gte_prediction`, **CatAL**) [3]. The baryon asymmetry $\eta_B = 6.109 \times 10^{-10}$ is derived via the FKTT (Fermi-Kink Top-Top) baryogenesis mechanism, machine-certified (`kink_top_coupling_eq_eps_FN`, `phi_md1_kink_bps_saturation`, **CatAL**) [3].

Table 1: Charged-fermion mass predictions vs. PDG 2024. All GTE predictions are derived from the InformationMassTransformer pipeline with the Elegant Kernel. The tau mass is the single anchor fixed by the Self-Consistency Condition; all other masses are outputs. RMS = 0.295% across all nine.

Fermion	PDG 2024	GTE prediction	σ	Cert.	Falsification threshold
e	0.5110 MeV	0.511 MeV	0.00	CatAL	— (anchor)
μ	105.66 MeV	105.66 MeV	0.00	CatAL	— (anchor)
τ	1776.86 MeV	1776.86 MeV	0.00	anchor	—
u	2.16 MeV	2.23 MeV	< 0.1	CatA	outside [2.0, 2.5] MeV
d	4.67 MeV	4.55 MeV	< 0.2	CatA	outside [4.0, 5.5] MeV
s	93.4 MeV	93.0 MeV	< 0.1	CatA	outside [85, 100] MeV
c	1275 MeV	1277 MeV	< 0.1	CatA	outside [1200, 1350] MeV
b	4183 MeV	4184 MeV	< 0.04	CatAD	outside [4100, 4300] MeV
t	172570 MeV	172610 MeV	< 0.03	CatA	outside [170, 175] GeV
Pion $\pi^\pm$	139.5703 MeV	139.57 MeV	0.00	CatAL	outside [139.0, 140.1] MeV

Table 2: Mixing-sector predictions vs. PDG 2024 and NuFIT 6.0 IC24 NH.

Observable	GTE prediction	PDG / NuFIT	σ	Cert.
Weinberg angle $\sin^2\theta_W$	0.23129 (two-loop)	0.23121 ± 0.00004	+0.038	CatAD
CKM CP phase δ_{CKM}	68.51°	$68.5^\circ \pm 2.5^\circ$	0.00	CatAL
PMNS CP phase δ_{CP}	$4/7 \times 360^\circ = 205.71^\circ$	$212^\circ \pm 26^\circ$	-0.15	CatA
PMNS solar $\sin^2\theta_{12}$	$4/13 = 0.3077$	0.308 ± 0.012	-0.02	CatAL
PMNS atm $\sin^2\theta_{23}$	$19/42 = 0.4524$	0.470 ± 0.015	-1.0	CatAL
PMNS reactor $\sin\theta_{13}$	$11/73 = 0.1507$	0.1489 ± 0.004	+0.23	CatAL

Table 3: Cosmological predictions vs. Planck 2018 and other measurements.

Observable	GTE prediction	Measurement	σ	Cert.
CMB spectral tilt n_s	$1 - \frac{\ln 2}{2\pi^2} = 0.96488$	0.9649 ± 0.0044	-0.005	CatAL
Baryon asymmetry η_B	6.109×10^{-10}	6.10×10^{-10}	+0.15	CatAL
Dark matter relic $\Omega_{\text{DM}} h^2$	0.11994	0.12011 ± 0.0012	-0.15%	CatAD
Ω_Λ (bracket)	$[3\pi/14, 0.6899]$	0.6889 ± 0.0056	in bracket	CatAD
Primordial tensor ratio r	= 0 exactly	< 0.036	consistent	CatA
Dark energy w	= -1 exactly	-1.006 ± 0.045	consistent	CatAD
Neutrino mass sum Σm_ν	59.4 meV	< 120 meV	consistent	CatA
Normal mass ordering	NH only	NH preferred	consistent	CatAD

2.4 Sector 4: Constants and Structure

Table 4: Structural and constant predictions.

Observable	GTE prediction	Value / status	σ	Cert.
Fine-structure constant α^{-1}	$2^7 + 3^2 = 137$	137.036	$\sim 0.026\%$	CatA
Newton’s constant G_N	m_τ^2/M_{Pl}^2	0.040% from PDG	~ 0.04	CatAD
Strong CP angle θ_{QCD}	$= 0$ exactly (F_{21})	$< 10^{-10}$	consistent	CatAL
Spacetime dimensions	$3 + 1$ (forced)	$3 + 1$ observed	exact	CatAD
EFT threshold Λ_{GTE}	≈ 2.03 GeV	near hadronic scale	target	CatAD

2.5 Sector 5: Dark Sector and New Physics

Table 5: Dark sector and new physics predictions.

Observable	GTE prediction	Status	Cert.	Experiment
Dark lepton GTE-P7	211.9 MeV	unobserved (predicted)	CatAD	Belle II
Dark matter χ_1	0.54 MeV UFIDM	$\sigma_{\text{SI}} \sim 10^{-61} \text{ cm}^2$	CatAD	—
Dark photon A'	does not exist	null searches confirmed	Lean	NA64, Belle II
Dim-4 proton decay	forbidden	$\tau_p > 1.6 \times 10^{34} \text{ yr}$	CatAL	Hyper-K, DUNE
Axion / ALP	does not exist	null searches confirmed	CatAL	ADMX, CASPER

3 The Strongest Confirmations

Among the 40+ quantitative GTE predictions, the following are the most precisely confirmed — predictions the framework was required to satisfy and did.

4 Current Tensions: An Honest Assessment

Scientific honesty requires presenting tensions clearly, not minimizing them.

Current tensions in the GTE prediction register

- **PMNS atmospheric mixing angle θ_{23} :** GTE predicts $\sin^2 \theta_{23} = 19/42 = 0.4524$ (**CatAL**). NuFIT 6.0 IC24 NH gives 0.470 ± 0.015 , which is -1.0σ from the GTE value. This is within 1σ and therefore not a confirmed discrepancy, but it is the most notable tension in the mixing sector. DUNE and Hyper-Kamiokande will resolve this to $< 0.5\sigma$ precision.
- **Cosmological constant (Route 2):** The GTE Ω_Λ upper bracket bound 0.6899 is a bound, not a standalone prediction. The Planck 2018 value 0.6889 ± 0.0056 lies 0.18σ below this bound. The lower bound $3\pi/14 \approx 0.6731$ is satisfied with $\sim 2.8\sigma$ margin. The bracket is **CatAD** but is presented as a bracket, not a point prediction; the theory does not yet predict the exact value of Ω_Λ within the bracket.
- **Hubble tension:** GTE predicts $H_0 = 67.95 \text{ km/s/Mpc}$ (**CatA**), consistent with Planck 2018 ($67.4 \pm 0.5 \text{ km/s/Mpc}$, $+1.1\sigma$) but in tension with local distance ladder measurements ($73.0 \pm 1.0 \text{ km/s/Mpc}$, $\sim 5\sigma$). GTE does not resolve the Hubble

tension — it predicts a Planck-consistent value and does not have a mechanism for a higher H_0 .

- **Strong coupling $\alpha_s(\Lambda_{\text{GTE}})$:** GTE predicts $e^2(\Lambda_{\text{GTE}}) = 7/2$ (physically normalized), which matches PDG α_s to within +7.4% at one loop (CatAD consistency). The residual offset after the precision kink form-factor measurement (+1.3–1.5 σ) is assigned to two-loop matching effects. An exact-grade upgrade requires the two-loop matching calculation.

None of these tensions constitutes a confirmed falsification. Each has a named pathway to resolution (a specific calculation or measurement).

5 Clean Falsification Criteria

The following is the definitive list of measurements that would falsify GTE at $> 3\sigma$ confidence, with no escape route, assuming the experiment is correctly calibrated.

Table 6: Clean falsification criteria: measurements that would definitively refute GTE as currently formulated. Each row gives the GTE prediction, the measurement precision threshold required for a $> 3\sigma$ refutation, the primary experiment, and the expected timeline.

Observable	GTE prediction	Falsifying measurement	Experiment	Timeline
CMB spectral tilt n_s	0.96488	$n_s > 0.98$ or < 0.95 at $> 3\sigma$	CMB-S4, Simons Obs.	2028+
Baryon asymmetry η_B	6.109×10^{-10}	outside $[5.9, 6.4] \times 10^{-10}$ at $> 3\sigma$	CMB-S4, BBN	2028+
CKM CP phase δ_{CKM}	68.51°	outside $[60^\circ, 77^\circ]$ at $> 3\sigma$	Belle II, LHCb	2027+
Weinberg angle $\sin^2\theta_W$	0.23129	outside $[0.228, 0.234]$ at $> 3\sigma$	HL-LHC, FCC-ee	2035+
Strong CP angle θ_{QCD}	$= 0$ exactly	any nEDM establishing $\theta \neq 0$ at $> 3\sigma$	nEDM, ILL	ongoing
PMNS $\delta_{\text{CP}}^{\text{PMNS}}$	205.71°	outside $[175^\circ, 240^\circ]$ at $> 3\sigma$	DUNE, Hyper-K	2030+
Δm_{21}^2 (solar)	$7.37 \times 10^{-5} \text{ eV}^2$	outside $[7.30, 7.44] \times 10^{-5}$ at $> 3\sigma$	JUNO	2025+
Neutrino mass sum	59.4 meV	$< 40 \text{ meV}$ or $> 80 \text{ meV}$ at $> 3\sigma$	CMB-S4, Euclid	2030+

Observable	GTE prediction		Falsifying measure- ment	Experiment	Timeline
GTE-P7 211.9 MeV	at	resonance exists	no neu- tral state in [182, 243] MeV (the pre- registered $\pm 14\%$ win- dow) in monophoton	Belle II	2031
Proton decay (dim-4)	forbidden		any verified dim-4 decay event	Hyper-K, DUNE	2035+
Dark photon A'	does not exist		any dark pho- ton discovery	NA64, Belle II	ongoing
Axion / ALP	does not exist		any confirmed axion signal	ADMX, CASPER	ongoing
Primordial tensors r	$r = 0$ exactly		$r > 0.01$ at $> 3\sigma$	CMB-S4, LiteBIRD	2028+
Inverted neutrino hi- erarchy	NH only		definitive IH determination	DUNE, Hyper-K	2030+

6 How to Use This Tutorial: A Guide for Each Community

6.1 For CMB Observers

The GTE predictions with the most immediate relevance to CMB science are: $n_s = 0.96488$ (already -0.005σ from Planck; CMB-S4 will test to $\sigma(n_s) \approx 0.003$), $r = 0$ (LiteBIRD reaches $\sigma(r) \approx 0.001$), and $\eta_B = 6.109 \times 10^{-10}$ (CMB-S4 will reduce the η_B uncertainty by a factor of ~ 3 , making the $+0.15\sigma$ a direct comparison).

The most decisive near-term CMB test is n_s : a confirmed $n_s > 0.975$ or $n_s < 0.955$ at $> 3\sigma$ from CMB-S4 would refute GTE’s defect-cosmology mechanism.

6.2 For Collider Physicists

The primary collider targets are:

- **Belle II:** GTE-P7 at 211.9 MeV (highest-priority new-physics signal), the dark-photon null (any discovery falsifies), and the dark lepton χ_2 at 24.5 MeV.
- **Hyper-Kamiokande / DUNE:** Proton stability (dimension-4 decay forbidden; dimension-6 decay via standard GUT mechanism not forbidden).
- **LHC / HL-LHC:** Dark lepton χ_3 at 3.60 GeV (mono-jet or Higgs invisible); the EFT threshold near $\Lambda_{\text{GTE}} \approx 2$ GeV.

6.3 For Dark Matter Hunters

GTE predicts UFIDM (ultra-faint interacting dark matter) at $m_{\chi_1} = 0.54$ MeV with $\sigma_{\text{SI}} \sim 10^{-61} \text{ cm}^2$ — invisible to WIMP-style direct detection. Any positive signal from WIMP searches (XENONnT, LZ, PandaX) at any mass above the neutrino floor would falsify the GTE dark-sector structure if the signal has the WIMP cross-section profile.

The GTE dark matter candidate can in principle be detected through its $SU(3)_{\text{dark}}$ interactions via stellar cooling constraints and MeV-scale astrophysical searches.

6.4 For Lattice QCD

GTE’s pion mass prediction $m_\pi = 139.57$ MeV and the GOR formula as a theorem are directly testable by lattice QCD. The prediction uses $m_u = 2.23$ MeV and $m_d = 4.55$ MeV as inputs; improved lattice determinations of the light quark masses will sharpen the comparison.

The kink cell count target $\xi_{\text{kink}} = 7$ cells (from the physical-point selection $a = \hbar c/\Lambda_{\text{GTE}}$) is a substrate Monte Carlo target: a lattice calculation of the kink correlation length at the GTE-derived lattice spacing should find $\xi_{\text{kink}} = 7$ (band $6.78\text{--}7.23 \pm 0.54$).

6.5 For Lean Formalization

The GTE Lean 4 library ([ugp-lean](#)) currently has 308 canonical modules with zero sorry, covering the key derivations. Open Lean targets that would upgrade predictions from **CatA** to **CatAL**:

- The sech-overlap integral bound: upgrade η_B to full **CatAL**(currently conditional on two **CatA** axioms for $I(5)$ and $I(11)$).
- The λ_1 -simplicity part of the Perron-Frobenius theorem for the spin-7 transfer matrix (awaiting upstream Mathlib support).
- The PMNS CP phase $\delta_{\text{CP}}^{\text{PMNS}} = 4/7 \times 360^\circ$ (currently **CatA**; orbital derivation exists, Lean formalization pending).

7 The Physical Incompleteness Theorem

Physical Incompleteness Theorem (**CatAD**)

GTE proves that the exact value of the cosmological constant Ω_Λ is a non-computable real number of Turing degree $0'$ (a member of $\Delta_2^0 \setminus \Delta_1^0$). No computable algorithm can determine the exact realized value; only computable bounds are accessible.

This means no experiment, however precise, can fully determine Ω_Λ within the GTE framework. The two bracket endpoints ($3\pi/14$ and 0.6899) are computable bounds on the non-computable realized value. This is analogous to Gödel’s incompleteness theorem applied to cosmological measurement.

This is arguably the single most transformative experimental probe of the GTE framework: if the Physical Incompleteness Theorem is correct, precision cosmological measurements will converge on Ω_Λ within the bracket without ever closing it to a point. Observations that consistently find $\Omega_\Lambda \in [0.673, 0.690]$ across independent methods would support this.

8 Summary: The GTE Falsifiability Profile

Key facts about GTE falsifiability

- GTE has **zero free parameters**: every numerical prediction is derived, not fitted.
- **40+ quantitative predictions** have been compared to PDG 2024, NuFIT 6.0, and Planck 2018. Approximately 37 agree within 1σ or match exactly; the remainder have named resolution pathways.

- The **strongest near-term tests** are: Belle II (GTE-P7 at 211.9 MeV), JUNO (Δm_{21}^2), CMB-S4 (n_s and r), and Hyper-Kamiokande (proton stability).
- The **null predictions** (no axion, no dark photon, no dim-4 proton decay, no SUSY particles) are confirmed by current searches and provide ongoing positive confirmation of the framework.
- Every new precision measurement from any frontier experiment is simultaneously a test of GTE, because the framework has no parameter room to absorb discrepancies.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P48 — The Complete GTE Framework** (main synthesis monograph; Appendix contains the complete predictions and falsification table):
doi.org/10.5281/zenodo.20560550
- **P47 — Cosmological Predictions of the GTE/ Φ_{MDL} Framework** (CMB, baryon asymmetry, cosmological constant):
doi.org/10.5281/zenodo.20465803
- **P01 — Standard Model Parameters from the UGP** (all nine charged-fermion masses and the mixing sector):
doi.org/10.5281/zenodo.20168787
- **P50 — The Spin-7 Lattice Model** (the kink cell count substrate target and spectral analysis):
doi.org/10.5281/zenodo.20603066

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

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